Gaussian context sampling for imbalanced regression with tabular in-context learners

TODO: authors, affiliation

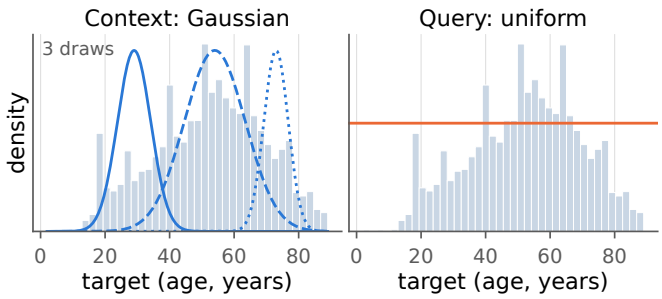


Figure 1: Context rows are weighted by a Gaussian over the target, with centre and width redrawn each episode; queries stay uniform.

Abstract

1 Introduction

2 Method

2.1 Context and query sampling

Each episode draws a centre and a width,

$$\begin{aligned} \mu &= Q_y(u), & u &\sim U(0.05, 0.95), \\ \sigma &= \mathrm{sd}(y)\, e^{v}, & v &\sim U(\log 0.15, \log 1.5), \end{aligned} \tag{1}$$

and weights each candidate context row by $w_i \propto \exp\!\big(-\tfrac{1}{2}((y_i-\mu)/\sigma)^2\big)$.

2.2 Two regimes

2.3 ECG encoder

3 Experiments

3.1 Setup

3.2 Results

3.3 The trade-off

4 Conclusion and limitations

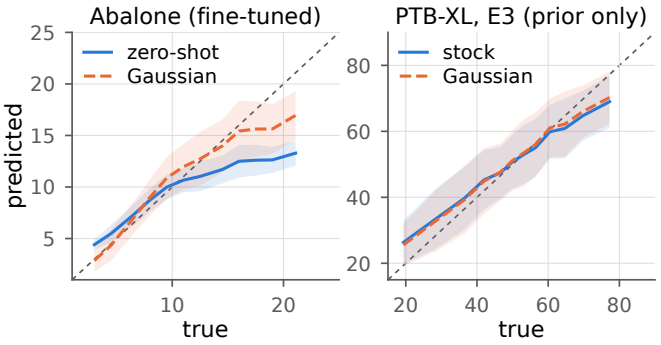


Figure 2: Binned predictions against truth, ±1 sd. TODO: state in the caption which arms and whether predictions are seed-averaged.

Table 1: TabICL, synthetic-prior continued pre-training only. No model has seen any evaluation dataset. Mean ± sd over 5 seeds; lower is better.

	Many	Medium	Few	Overall	GM
Abalone					
zero-shot	1.146 ± 0.081	2.595 ± 0.031	6.219 ± 0.030	2.956 ± 0.038	1.649 ± 0.085
uniform context	1.148 ± 0.082	$\mathbf{2.578 \pm 0.031}$	$\mathbf{6.189 \pm 0.031}$	$\mathbf{2.943 \pm 0.039}$	$\mathbf{1.645 \pm 0.077}$
target-balanced context	1.163 ± 0.054	3.226 ± 0.027	7.233 ± 0.032	3.432 ± 0.023	1.989 ± 0.064
Gaussian context	$\mathbf{1.128 \pm 0.087}$	2.618 ± 0.024	6.385 ± 0.037	3.000 ± 0.028	1.665 ± 0.075
Communities					
zero-shot	0.0587 ± 0.0013	$\mathbf{0.1680 \pm 0.0054}$	0.2552 ± 0.0083	0.1598 ± 0.0040	0.0898 ± 0.0068
uniform context	0.0585 ± 0.0014	0.1683 ± 0.0055	$\mathbf{0.2549 \pm 0.0082}$	$\mathbf{0.1598 \pm 0.0040}$	$\mathbf{0.0896 \pm 0.0072}$
target-balanced context	$\mathbf{0.0514 \pm 0.0023}$	0.2126 ± 0.0059	0.3552 ± 0.0024	0.2047 ± 0.0020	0.1135 ± 0.0025
Gaussian context	0.0577 ± 0.0015	0.1737 ± 0.0066	0.2854 ± 0.0037	0.1708 ± 0.0031	0.0937 ± 0.0042
Wine Quality White					
zero-shot	$\mathbf{0.335 \pm 0.008}$	$\mathbf{1.195 \pm 0.060}$	$\mathbf{2.453 \pm 0.024}$	$\mathbf{0.785 \pm 0.020}$	$\mathbf{0.256 \pm 0.008}$
uniform context	0.337 ± 0.009	1.199 ± 0.059	2.456 ± 0.025	0.787 ± 0.019	0.267 ± 0.009
target-balanced context	0.345 ± 0.009	1.275 ± 0.055	2.535 ± 0.019	0.825 ± 0.019	0.285 ± 0.013
Gaussian context	0.349 ± 0.011	1.280 ± 0.051	2.517 ± 0.022	0.828 ± 0.018	0.353 ± 0.010

Table 2: TabPFN-2.5, per-dataset fine-tuning. Learning rate selected on validation. The target-balanced control is specified on Abalone only. Its row almost reproduces zero-shot because it never improved validation RMSE over the stock model in 14 of 15 (seed, learning rate) runs, so early stopping returned the un-fine-tuned model.

	Many	Medium	Few	Overall
Abalone				
zero-shot	$\mathbf{1.170 \pm 0.077}$	2.569 ± 0.028	6.163 ± 0.040	2.942 ± 0.029
uniform fine-tuning	1.208 ± 0.086	2.357 ± 0.105	5.767 ± 0.217	2.786 ± 0.108
Gaussian fine-tuning	1.584 ± 0.145	$\mathbf{1.392 \pm 0.094}$	$\mathbf{4.127 \pm 0.151}$	$\mathbf{2.202 \pm 0.062}$
target-balanced fine-tuning	1.176 ± 0.077	2.567 ± 0.027	6.139 ± 0.063	2.938 ± 0.030
Communities				
zero-shot	$\mathbf{0.0592 \pm 0.0015}$	0.1677 ± 0.0049	0.2577 ± 0.0053	0.1606 ± 0.0035
uniform fine-tuning	0.0597 ± 0.0016	0.1678 ± 0.0051	0.2538 ± 0.0062	0.1596 ± 0.0026
Gaussian fine-tuning	0.0646 ± 0.0031	$\mathbf{0.1617 \pm 0.0042}$	$\mathbf{0.2348 \pm 0.0012}$	$\mathbf{0.1530 \pm 0.0020}$
Wine Quality White				
zero-shot	0.427 ± 0.013	1.432 ± 0.022	2.455 ± 0.027	0.926 ± 0.011
uniform fine-tuning	$\mathbf{0.370 \pm 0.019}$	1.295 ± 0.023	2.375 ± 0.025	0.837 ± 0.011
Gaussian fine-tuning	0.391 ± 0.038	$\mathbf{1.138 \pm 0.055}$	$\mathbf{2.250 \pm 0.104}$	$\mathbf{0.783 \pm 0.031}$

Table 3: PTB-XL age prediction, mean absolute error in years, on the official test split. Band-macro is the unweighted mean of the three band errors.

	<30	30–60	≥60	Natural	Band-macro	GM
ResNet (vanilla)	8.603 ± 0.312	$\mathbf{6.726 \pm 0.177}$	8.334 ± 0.553	7.483 ± 0.087	7.888 ± 0.129	4.869 ± 0.080
ResNet + RankSim + ConR	8.399 ± 0.504	6.847 ± 0.174	8.425 ± 0.317	7.551 ± 0.138	7.890 ± 0.123	4.905 ± 0.110
E2 + stock TabPFN-2.5	7.659 ± 0.013	7.246 ± 0.006	7.628 ± 0.010	7.421 ± 0.003	7.511 ± 0.006	4.728 ± 0.015
E2 + Gaussian TabPFN-2.5	7.663 ± 0.017	7.244 ± 0.006	7.629 ± 0.010	7.421 ± 0.003	7.512 ± 0.007	4.731 ± 0.014
E3 + stock TabICL	7.600 ± 0.020	7.152 ± 0.005	7.601 ± 0.006	$\mathbf{7.353 \pm 0.002}$	7.451 ± 0.005	$\mathbf{4.716 \pm 0.038}$
E3 + Gaussian TabICL	$\mathbf{7.059 \pm 0.053}$	7.849 ± 0.061	$\mathbf{7.342 \pm 0.053}$	7.585 ± 0.017	$\mathbf{7.417 \pm 0.017}$	4.903 ± 0.058

Table 4: Encoder training against prediction head. The gap between the two column groups is the contamination in the balanced protocol, not a property of the method.

	Natural (official split)		Balanced (DIR split)	
Encoder	stock head	Gaussian head	stock head	Gaussian head
E1 (supervised L1)	7.428 ± 0.008	7.654 ± 0.041	7.330 ± 0.067	7.162 ± 0.070
E2 (through TabPFN)	7.428 ± 0.004	7.648 ± 0.015	7.258 ± 0.090	7.108 ± 0.080
E3 (through stock TabICL)	$\mathbf{7.353 \pm 0.002}$	$\mathbf{7.585 \pm 0.017}$	$\mathbf{7.152 \pm 0.090}$	$\mathbf{7.042 \pm 0.120}$
E4 (through Gaussian TabICL)	7.469 ± 0.004	7.617 ± 0.007	7.384 ± 0.080	7.392 ± 0.108